The Gender Wage Gap: Triple penalty? Woman, mother and informal

Jonathan Garcia[*]

15 de octubre de 2025

Abstract

Family formation, child-rearing, and their implications for women's labor market decisions have recently been studied as potential "unexplained factors" that contribute to the gender pay gap. Opposite to most of the literature available, I study the short-term impacts of motherhood, marriage, and labor market interruptions in Nicaragua, a country with high female labor force participation, shared maternity leave costs, and a high informality. In this paper, I provide evidence of unexplained wage differences by gender due to motherhood and employment in the informal sector. I use labor market transitions to assess a potential mechanism, mothers preferring informal jobs due to flexible working hours. I find that formal jobs represent an absorbing and a prefereble job status for mothers, even more than for men (with and without children). Likewise, mothers show higher transition rates from unemployment and inactivity to informal jobs. These results suggest that mothers are even keener to work, contrary to what an employer (due to prejudice or statistically bias) may assume and that the selection into informal or "flexible jobs" is influenced by the lack of better jobs and not by voluntary choice.

JEL Codes: J13, J16, J46, J69, J71.
Keywords: gender gap, motherhood penalty, informality, labor transitions.

[*]Humboldt University of Berlin, IGZ. jonathan.garcia@hu-berlin.de.
I am thankful for the comments and revisions from Rony Rodríguez-Ramírez, Henrike Sternberg, Juliane Zenker, and the evaluation committee of the Economy, Finance and Development Award of Nicaragua's Central Bank. I am also grateful to Guerson Salgado for his support with the translation to Spanish. The paper was written during my time as EPOS Scholar, funded by the German Academic Exchange Service, or DAAD.

1. Introduction

Whether looked upon from a normative or an empirical point of view, gender discrimination is harmful to the economy, to society, and most importantly, to the largest affected group, women. Recent economic growth and development have increased women's access to education and health, yet, the gender wage gap in the developing and developed world persists. The International Labor Organization (ILO) forecasts that, if continued at the current pace, as many as 270 years are needed to close this gap (International Labour Organization, 2018). However, in the meantime, the uneven returns to education and productivity continue to represent output and welfare losses for women and their households.

Recently, substantial efforts and innovative research have been conducted to understand the "unexplained" differentials on earnings by gender. Part of this research has concentrated on understanding how the trade-off between time and earnings imposed by the child-rearing process affects the gender pay gap. This literature shows that both parents face the decision of working a full-time job or opting for a flexible job and, in many cases, exiting the labor market. Literature also shows that mothers and no father are usually the ones who choose to work less or move out of the labor force, generating a lasting negative effect on mothers' earnings throughout their stay in the labor market (Berniell et al., 2021; del Boca et al., 2013; Goldin, 2014; Kleven et al., 2019).

Most empirical research has been conducted in developed countries, especially the United States and Europe, where welfare states, family policies and labor market structures differ substantially from those in the developing world. In Latin America, the limited offer of "quality jobs" may play a role in parents' decision-making for opting for more flexible options during their children's upbringing. Flexible jobs may be mostly jobs in the informal sector, which comprises a large share of the region's labor supply, and it mostly entails uninsured and low-paid jobs. Furthermore, differences in the parental leave policies for mothers and fathers may increase mothers' pay gaps via employer discrimination. For instance, in Nicaragua, while mothers are entitled to 12 or up to 14 weeks of leave when given birth, fathers are only allowed five days according to the current labor code.[1].

Nicaragua, with a high female labor force participation (70 percent[2]), a high level of infor-

[1]Nicaragua's Labor Code, reforms and amendments. Law 185

[2]Author estimations based on the Continues Household Survey, IV quarter of 2012.

mality (80 percent) and family policies such as shared parental leave cost between employers (40 percent) and the National Institute of Social Security (60 percent), offers an appropriate and interesting field to investigate the gender wage gap as well as the implications of parenthood and informality. Furthermore, the country collects a quarterly labor force survey, the Continues Household Survey (ECH by its Spanish acronyms), in a panel of more than 3,000 households representative at the urban and rural area that allows studying labor market dynamics and control for invariant characteristics at the individual level.

While identifying the existence of a gender wage gap and its changes across different groups, this study aims to shed light on the possible triple burden of discrimination that women could face in Nicaragua's labor market due to motherhood and their insertion in the informal sector. Thus, I study the gender wage gap, first via estimating wage equations à la Mincer controlling for individual, household, supply-side and unobserved characteristics and second via estimating Oaxaca-Blinder decomposition across different sample groups (women vs. men, mothers vs. fathers, mothers vs. fathers in the informal sector) to identify wage differentials after controlling for sample self-selection.

I find evidence of a gender wage gap between men and women of around 19 percent and that this difference increase while comparing mothers and fathers in the formal sector only. The gap between mothers and fathers in the informal sector seems to be lower than in the formal sector. However, this result is driven by differences in the endowments, which fade out after controlling for self-selection. Thus, the gender gap for mothers in the informal sector is larger than that for mothers in the formal economy and the entire group of women, suggesting that women in informality may face additional penalties for their mother and informal conditions.

Aiming to provide insights into the nature of this wage differential, I assess one possible explanation for the pay gap; the belief that women, especially mothers, prefer more flexible jobs than full-time jobs due to the household's reproductive tasks' burden. Hence, I analyze labor market mobility patterns for men, women, fathers and mothers, focusing on identifying differences in their permanence in formal jobs or, equivalently, transitions towards more flexible jobs, in this case, represented as informal jobs. I use Markov Transition Matrices and find that the permanence rate in formal jobs is higher for mothers than for only men and fathers.

Moreover, transitions out of inactivity and unemployment increased for mothers compared to mothers without children, men and fathers. Women increase their probability of entering informal jobs status (own-account and salaried worker) when they become mothers.

The remaining part of the paper is structured as follows. Section II provides an extensive review of the existing literature on the gender wage gap, its measurement, and the current factors that are believed to explain it, with a major focus on the research examining the "unexplained" factors. In section III, I explain the data and primary methodologies implemented in the study. In section IV, I provide a brief description of the Nicaraguan labor market. Section V presents the main estimations' results and the final section VI concludes and provides policy recommendations.

2. Gender wage gap

The gender wage gap refers to the systematic differences in the outcomes that men and women, identical as regards their productive ability, achieve in the labor market because of specific non-productive characteristics. These differences appear in the percentages of men and women in the labor force, the types of occupations they choose, and the difference in the average incomes of men and women (Goldin, 2008).

The total wage differential between men and women can be decomposed into an explained part due to differences in characteristics and an unexplained residual component. The role of gender differences in this unexplained residual is often referred to as the discrimination effect. Empirical research on gender pay gaps has traditionally focused on the role of gender-specific factors, particularly gender differences in qualifications and differences in the treatment of otherwise equally qualified male and female workers (i.e., labor market discrimination) (Blau & Kahn, 1992; Weichselbaumer & Winter-Ebmer, 2005)

The economic analysis of discrimination can be traced back to the initial work on the economics of Discrimination by Becker (1985) and the vast literature that followed. Becker's theory of labor market discrimination is based on the taste-based discrimination concept. Applied to gender differences, this concept refers to the notion of gender prejudice and economic outcomes. Becker monetizes this prejudice and determines its different sources, classified as employer discrimination, employee discrimination and customer discrimination (Borjas, 2016).

In addition, or even in the absence of prejudice-based discrimination, differences in skills or productivity are being attached to specific genders. Employers have imperfect information about individual productivity and only know group-level average productivity, e.g., women. Thus, the perceived productivity of a worker by the employee leads to unexplained differences. The notion of this theory is based on the initial work by Arrow (1973) and Phelps (1972), and it is known as statistical discrimination (Altonji & Blank, 1999).

The gender wage gap, and the factors it is determined by, have been subject to constant changes across time, at a similar pace at which the labor market changes. Up to date, traditional factors such as union status, race, migration status or religion, and especially human capital-related variables, i.e., educational attainment and experience, explain very little of the differential earnings among men and women (Blau & Kahn, 2017; Weichselbaumer & Winter-Ebmer, 2005). Recently, the literature has increasingly analyzed other factors due to newly observed phenomenons or improvements concerning the methods used to study the contributions of these factors to the wage differentials.

Over the improvements of the past century, the gender gap in education has almost vanished and the one in educational attainment has even been reversed in developed and developing countries (Becker et al., 2010; Gaddis & Klasen, 2013; Goldin, 2006). This trend is also observed in the case of experience level in the labor market. In the United States, by 2011, women were three percent more likely to achieve an advanced degree than men, and the years of full-time experience of men (17.8 years) were only 1.4 years higher than women (Blau & Kahn, 2017). Thus, instead of explaining wage differentials, education level and experience nowadays rather represent a reducing factor of the gender wage gap.

2.1. Labor market interruptions

Labor market attachment differences have been at the center of the debate in explaining the gender wage gap (Mincer & Polachek, 1974). Women's labor market interruptions, mainly due to family formation, i.e., marriage and childbirth, still account for a significant percentage of wage differentials among men and women (World Bank, 2012). For instance, Goldin (2014) found that job interruptions for MBA and finance professionals in the United States contribute to up to 30 percent of the gender gap in earnings.

Even though family formation arrangements have changed over time, differences are still observed across countries and regions derived primarily from the formal and informal institutions prevalent in their countries. In Argentina, Brazil, Ghana, Mexico, Serbia, and Thailand, differences between men and women in the time use for reproductive work affect their likelihood to transit to "good jobs" negatively and increase their probabilities of being in the informal self-employed sector or the inactive labor force (Bosch & Maloney, 2010).

The magnitude of the contribution of job interruptions depends on their length and measurement. Empirical evidence finds support for positive impacts of short leaves, but negative effects for long leaves on women's wages (Blau & Kahn, 2000; Ruhm, 1998; Waldfogel, 1998).Nonetheless, most of the empirical literature encounters the issue of lacking an appropriate way to measure job interruptions and their length, thus, leading to imperfect proxies. Nordman et al. (2016) provide evidence for the possibly misleading results of the use of proxies while exploiting a novel database from Madagascar that allowed them to compare the contribution of actual and potential experience (proxy). They find that the latter underestimates the contribution of job interruptions to the gender wage differentials.

Consequently, differences in job tenure that are usually in favor of male are found to have significant contributions to the widening of the earning gaps throughout their professional careers (Kleven et al., 2019) with heterogeneous magnitudes across regions, from around 1 percent of the wage's determinants for females in Europe and up to 10 percent in developing countries (Weichselbaumer & Winter-Ebmer, 2005).

2.2. Motherhood penalty

Empirical evidence suggests that, in addition to the impact on labor market attachment, gender roles and family formation decisions might affect women's relative wages (Blau & Kahn, 2017). This phenomenon was initially called the family gap, representing the difference in pay between women with children and women without children (Waldfogel, 1998). Currently, it is better known in the literature as the motherhood penalty.

Budig and England (2001) use a fixed-effects model for the United Status aimed at measuring the magnitude of the motherhood penalty and obtained results that show a wage penalty of 7 percent per child. It also found that the penalties are larger for married women than for unmarried women and that women with (more) children have fewer years of job experience, as well as observing that, after controlling for experience, a penalty of 5 percent per child remains.

Using data on seven countries with different welfare states, Sigle-Rushton and Waldfogel (2007) found that gaps in family income are smallest in the Nordic countries, intermediate in the Anglo-American countries, and largest in the continental European countries. Thus, the effect of motherhood differs according to the country-type welfare state.

Kleven et al. (2019) used Danish data and find that the arrival of a child creates a gender gap in earnings of around 20 percent driven mainly by job interruptions, hours of work, and wage rates, all affecting mothers, but not fathers. The gender gaps generated are very stable over time, and women show no sign of labor market recovery even ten years after the first child. It also found that child penalty is transmitted through generations, to daughters and not to sons, via environmental influences in the formation of women's preferences over family and career.

Estimating and establishing a causal effect of the motherhood penalty remains controversial due to several factors such as women's self-selection, i.e. women with lower wages have lower costs of children, inter alia. Several factors arise from the literature as a possible explanation of the motherhood penalty, among those, employers' discrimination and the trade-off of working hours or higher wages for mother-friendly jobs (Budig & England, 2001).

Experimental evidence for discrimination has aimed to assess the existence of such and the possible determinants. Correll et al. (2007) conducted a laboratory experiment in which résumés of equally qualified same-sex job applicants, who differed only by parental status, were assessed. As a result, lower starting salaries were recommended for mothers, while evaluators did not penalize men for being fathers. Further, Benard and Correll (2010) attempted to examine whether mothers face discrimination in labor-market-type evaluations, even when they provide indisputable evidence that they are competent and committed to paid work. Their results suggest that female evaluators apply "normative discrimination," finding mothers less warm, less likable, and more interpersonally hostile than otherwise similar workers who are not mothers. Alternatively, Williams and Ceci (2015) did not find evidence of discrimination in the academic market and concluded that discrimination towards mothers could result from statistical discrimination due to employers' perception of productivity differences between mothers and non-mothers.

Gender differences in time use, especially in non-market responsibilities, impact women's labor market outcomes in several ways (Becker, 1985; Blau & Kahn, 2017). For instance, long hours that married women or mothers spend on reproductive activities could reduce the effort they put into their market jobs. Likewise, Hersch and Stratton (2002) find that housework, especially daily routine tasks such as cooking and cleaning, harms wages regardless of marital status in the United States. Also, it provides evidence that controlling for housework time increases the explained component of the gender wage gap by 14 percentage points. Cha and Weeden (2014) examined the role of an increase in the prevalence of long work hours and the gender wage gap during the 1979–2009 period. They find that the difference in hours worked increased the gender wage gap by about 10 percent of the total change over this period.

(Goldin, 2014) analyzes the impact of temporal flexibility or flexible jobs taken by women and mothers on the gender wage gap. Goldin refers to flexibility as a multitude of temporal matters, including the number of hours, precise times, predictability and ability to schedule one's own hours. For the United States case, Goldin found that women with children work 24 percent fewer hours per week than men or women without children. Her research focuses on the disproportionate rewards in some occupations or firms for working long hours or be available on call. She concludes that earnings have a nonlinear relationship concerning hours, and that this relationship has major implications in the corporate, financial, and legal worlds.

2.3. Informality

Labor markets in developing countries are characterized by high levels of informality. For instance, in Latin America, the so-called informal sector comprises a large share (30 to 70 percent) of the occupied labor force (Maloney, 2004). Jobs in the informal sector differ from formal jobs in dimensions such as weaker social protection, lower labor earnings, lesser career prospects, more flexibility and shorter working hours (Berniell et al., 2021). Thus, the main conclusions of the vast literature about the gender wage gap that have focused mainly on the United States and Europe might have different applications to countries with high informality rates.

Tansel et al. (2019) estimated the Oaxaca-Blinder decomposition of wage differentials between men and women by different social insurance coverage status in Turkey. In this context, it founds that in the covered sector, men's wages are about twofold higher than women's wages. For the uncovered wage workers, men's wages are near parity with those of women. These results suggest segmentation for men in the formal and informal sector and substantial discrimination for women in the covered private sector.

Similarly, Ruzik and Rokicka (2010) analyzed the gender wage gap in Poland's informal sector using quantile regressions, in this case formally register and unregister workers, finding that the inequality of earnings among unregistered women and men is more pronounced at the bottom tail of the earnings distribution. In the case of formal employees, inequality at the top of the distribution tends to be larger, confirming the existence of a 'glass ceiling'. It also proposes that a possible explanation of the results is the lack of minimum wage regulations in the informal market and the greater flexibility in agreement on wages in the higher quantiles.

Conversely, Yahmed (2018) studied how gender inequality differs across formal and informal workers in Brazil, finding that the raw gender wage gap is about the same in informal jobs and formal jobs. However, it also finds that this result is driven by the difference in the male and female selection processes. Hence, after controlling for observable characteristics, the adjusted gender wage gap is, on average, about 24 percent among formal employees and about 20 percent among informal employees.

Lastly, Berniell et al. (2021) further investigate the effect of gender differences by formalization status and the motherhood effect for Chile and other OECD developing countries, finding that motherhood produces a sizable reduction in labor earnings for Chilean mothers and that this reduction is associated with a lower labor force participation and a drop in the formal employment. Likewise, the study finds that Chile's motherhood penalty is smaller than in the US, but it is larger than in Denmark.

2.4. Measurement

Different methods have been developed to analyze discrimination empirically. The most common way to analyze discrimination based on gender is to compare male and female earnings holding productivity constant. One method is to include a sex dummy in the wage regression model only. However, the standard procedure to investigate differences in wages is by using decomposition methods (Fortin et al., 2011; Weichselbaumer & Winter-Ebmer, 2005).

Several methods have been developed since Oaxaca (1973) and Blinder (1973) seminal work. However, the so-called Oaxaca-Blinder decomposition continues to be the most frequently used approach. This procedure allows decomposing the wage differential between men and women into an explained part due to differences in characteristics and an unexplained residual (Fortin et al., 2011; Weichselbaumer & Winter-Ebmer, 2005).

2.5. The gender wage gap in Nicaragua

Few studies have analyzed the gender wage gap in Nicaragua. Weichselbaumer and Winter-Ebmer (2005) conduct a meta-analysis and conclude that a substantial part of the total wage gap can be attributed to differences in human capital, given their findings that women have lower endowments than men. Furthermore, Monroy (2008) estimates raw differences in income for male and female workers by occupation, finding an average difference of 20 percent. Likewise, after estimating raw differences in income by formalization status, it finds that the gap in the informal sector (18 percent) is higher than in the formal sector (7 percent).

Similarly, Programa de las Naciones Unidas para el Desarrollo (2014) report about the Nicaraguan labor market from a gender perspective uses Oaxaca-Blinder decompositions and finds that men's monthly income is higher than women's income in more than 30 percent for both rural and urban areas. Nonetheless, the point estimates and such a difference could be attributed to self-selection issues.

The present paper contributes to the literature on the gender wage gap in Nicaragua by implementing improved econometric techniques and applying them on a panel dataset to estimate wage differences while controlling for individual and family characteristics and possible self-selection issues. Furthermore, the survey's panel characteristic allows for the study of job interruptions on wages and the analysis of plausible channels of the motherhood penalty, particularly the likely selection of mothers into more flexible or informal jobs. Lastly, analyzing the gender pay gap after solving for self-selection might shed light on the shared maternity leave policy's particular effect while comparing formal and informal sectors. This policy can induce prejudice and discrimination from employers, and, therefore, investigating its implications is in the interest of policymakers.

3. Data and methodology

I use the Continues Household Survey (ECH) to estimate the models described in this paper. The ECH collects quarterly information on socioeconomic, and labor market conditions in a rotative panel with more than seven thousand households focused on the active labor force. It is representative of the rural and urban level and provides enough information on formal and informal workers.

The core source of information of the following estimations are the databases of the four quarters of 2012. The applied unit of analysis are individuals belonging to the labor force, containing people between 14 and 65 years old (80,863 observations). The lower value has been defined according to the minimum age used at a national level to define the labor force by the Central Bank of Nicaragua and INIDE (Banco Central de Nicaragua, 2009).

The main outcome variable, wages, refers to all the labor income of primary and secondary occupations, and it has been constructed based on the information collected by the survey. The survey does not include direct information about mothers. Hence, I code as a mother, a woman being the partner of the head of the household, or the head of the household when there are children. Fathers are coded in a similar fashion. Job interruptions are coded using the panel. The dummy for job interruptions takes a value of one when an individual in the previous quarter of the survey was unemployed, out of the labor force, or inactive and in the current survey has changed to any type of employment status.

Employment status has been analyzed according to the 19th International Conference of Labour Statisticians (ICLS) (International Labour Organization, 2013). as the characteristics of the ECH survey allow to do so. The Informal Sector Enterprise and job-based state classification have been measured according to the 17^{th} ICLS (International Labour Organization, 2013). Industry classification has been procced using United Nations (2008) and OECD (Elias, 1997).

The primary estimations applied in this study aim to provide evidence on the existence of a gender wage gap in Nicaragua, as well as to shed light on its magnitudes and possible drivers. Those estimations are: (i) determinants of monthly wages and (ii) a Oaxaca-Blinder decomposition, whereas both, (i) and (ii), also analyze gender differences by formalization status. Furthermore, to study a potential channel of the motherhood penalty, mother opting for "flexible jobs", I (iii) estimate transition matrices derived from the Markov transition process to provide evidence on women's and mothers' labor market mobility.

Wage equations à la Mincer are implemented for monthly earnings. The first estimation is a pooled Ordinary Least Squares (OLS) of a sample that includes male and female wages, intending to provide initial evidence of the impact of gender in wage determination, followed by separate models for males and females. Wage equations include individual, household and supply-side covariates and are expressed as follow:

$$Ln\ W_{it} = \beta_0 + \beta_1 X'_{it} + \beta_2 X'_{ht} + \beta_3 X'_{st} + \varepsilon_{it}, \tag{1}$$

where $Ln\ W$ represents the logarithm of wages for each individual i in time t. The main set of covariates includes variables at individual and household level. The vector of covariates X'_{it} includes job status according to ILO classification, age, age squared, area of residence, a married dummy, educational attainment, industry, a mother dummy, a father dummy, a dummy to proxy job interruption signaling those with previous quarter out of the labor force, inactive or unemployed, and a dummy for being currently in school or training. At the household levels, the covariates vector X'_{ht} includes the following controls: number of children, dummies for a household receiving remittances, pension and being beneficiary of public school programs, and household size. Lastly, vector X'_{st} represents dummy variables (fixed-effects) for the economic strata identified in the sampling frame and for each quarter of the survey to further control for socioeconomic characteristics or possible idiosyncratic shocks at the area of residence (strata) or in time (quarters).

Subsequently, I estimate the same wage equations using individual fixed effects aiming to partially control for unobserved heterogeneity, such as workers' ability that might confound previous OLS estimations (Angrist & Pischke, 2008). Wage equations with fixed-effects are as follow:

$$Ln\ W_{it} = \alpha_i + \gamma_t + \beta_0 X_{it} + \beta_1 X_{ht} + \beta_2 X_{st} + \varepsilon_{it}, \tag{2}$$

where α_i represents individuals fixed-effects and γ_t time fixed-effects.

Estimations of the gender wage gaps are conducted following the work of Blinder (1973) and Oaxaca (1973) who proposed a decomposition procedure that divides the wage differential between two groups into a part that is "explained" by group differences in productivity characteristics, such as education, and a residual part that cannot be accounted for by such differences in wage determinants. This "unexplained" part is often used as a measure for discrimination, but it also subsumes the effects of group differences in unobserved predictors (Jann, 2008).

Based on the initial wage equations, Bliender-Oaxaca decomposition can be expressed as follow:

$$\overline{W}_m - \overline{W}_f = (\overline{X}_m - \overline{W}_f)\hat{\beta}_m + (\hat{\beta}_m - \hat{\beta}_f)\overline{X}_f = E + U, \tag{3}$$

where $\overline{W}_m - \overline{W}_f$ denote the mean logarithms wages and control characteristics for each group. This difference equales two terms, a first term that stands for the effect of different productive characteristics $(\overline{X}_m - \overline{W}_f)\hat{\beta}_m$, and the second term represents the unexplained residual $(\hat{\beta}_m - \hat{\beta}_f)\overline{X}_f$, the differences in the estimated coefficients for both groups is usually called the discrimination effect (Weichselbaumer & Winter-Ebmer, 2005).

Initial wage decomposition is presented for a pooled sample of individuals analyzing men and their contractual 'women', followed by a decomposition analysis by parenthood and formalization status. These models intend to provide raw estimates of the wage differential and its components. Nonetheless, wages are only observed for individuals participating in the labor force causing potential self-selection issues. Therefore, in order to control for this issue, sample-correction procedures as proposed by (Heckman, 1979). This procedure entails the modelization of the labor force participation as a function of age, marital status and the number of children.

To investigate the probability of transiting among different job status, flexible and non-flexible status (approximated as informal and formal jobs), inter-quarterly transition matrices, are estimated to provide a general impression of the employment mobility patterns by gender and by parenthood status. The estimated transition matrices follow a stochastic and discrete process know as Markov process(Nichols, 2014). Transition matrices are determined as follows:

$$s_{t+1} = M_1 s_t \tag{4}$$

where M_1 represents a Markov matrix containing the probability of individual i of being in a particular row of the matrix (job status) in time $t+1$. Likewise, the probabilities in the Markov matrix are estimated as follows:

$$p_i(t, t+1) = Pr(s_{t+1}|s_t) \tag{5}$$

where $p_i(t, t+1)$ represents the conditional probability of finding an individual i en in a job status s in time $t+1$ given that the individual was in a particular job status S in time t. The sum of each row in the transition matrix is equal to 1, while the diagonal represents the permanence in the initial job status, depicting the most persisting or absorbing job status.

The transition matrices are estimated with an unbalanced panel of the ECH with data of the four quarters of 2012, for which methods to rectangularize the data were applied in the statistical software in order to obtain Markov transitions matrices. For these matrices, six employment status classifications were considered: formal-salaried workers, informal workers, self-employed/own-account workers, informal self-employed/own-account workers, unemployed, and out of the labor force or inactive population.[3]

4. Labor market in Nicaragua

In this section, a brief description of the Nicaraguan labor market is provided based on the Continuous Household Survey (ECH) of the last quarter of 2012. All descriptive statistics use expansion factors to enable the inference of the population.

The total survey sample is divided into 49 percent men and 51 percent women. Among those men in the survey, 63 percent belong to the working-age population and among women, this proportion represents 65 percent. The occupied population, as defined by the 19th ICLS, ascend to 90 percent of men and 71 percent of women in the working-age population. Despite

[3] Out of the labor force and inactive status were merged due to the small sample size of inactive population.

the differences in participation, informality rates among both groups are similar, 80 percent for men and 81 percent for women. Similarly, the share of the population living in the rural sector accounts for 44 percent of males and 41 percent of females.

On average, the observed monthly income for men is 3,346 local currency (córdobas), which represents US$142 at the time. For women, the average income is 2,824 local currency, which corresponds to US$120. Thus, women's income represents 84 percent of men's income. However, the average number of hours that a woman works (30 hours) represents 75 percent of the average hours worked by a man (41 hours).

Men and women present similar levels of educational attainment with regard to the completion of primary, secondary and advanced degrees. The share of Men with no formal education is 24 percent compared to 23 percent of women. Likewise, 11 percent of men completed primary school compared with 10 percent of women. Contrarily, eight percent of women completed secondary school compared with seven percent of men and 10 percent of females achieved an advanced degree compared with nine percent of males. Thus, the difference seems not to be significant; this might indicate that any pay gap is not attributed to education differences.

In terms of occupational segregation, the major differences are observed in the agriculture and fishing industry, where 45 percent of males are employed compared with only 21 percent of females. Similarly, the construction sector employs seven percent of the male labor force and only one percent of the female labor force. In contrast, the industry of trade, hotels and restaurants employs 38 percent of the female labor force and only 19 percent of the male labor force. Moreover, the industry of education, health and social protection hire 15 percent of the female labor force and 6 percent of the male labor force.

Furthermore, the survey data show at the macro-economic sector level that approximately 66 percent of women insert in the tertiary sector, especially in commerce, sales, and personal services, and usually as own-account workers, while this sector only employs 38 percent of the male labor force. Conversely, the primary sector contains 45 percent of the male labor force and 21 percent of the female labor force. The secondary sector shows more balanced participation, employing 17 percent of men and 13 percent of women in the labor force.

5. Evidence of a Gender Wage Gap

5.1. Determinants of wages

In order to inspect the impact of gender on wage determination and possible discrimination based on gender, I estimate wage equations initially using the pooled sample of male and females, and later, performing an estimation of wage determinants by gender while controlling for unobserved heterogeneity with fixed-effects to inspect possible drivers of those differences. I compared the fixed effects model with a random effect using the Hausman test, resulting in being the fixed-effects model the preferred one.

Table 5.1 shows the results from equation (1) and (2). In general, being female reduces the average net labor income by roughly 20 percent. Also, being married has a positive and significant effect on incomes of about 8 percent. The mother dummy is insignificant while being a father has a significant and positive effect of about 8 percent on the labor income. Further, having been unemployed, out of the labor force, or inactive in the previous quarter has a significant and negative effect on incomes, reducing it by roughly 30 percent. Other control variables such as age, age squared, job status, attending school and non-labor income dummies such as pension and public programs are significant and show directions and magnitudes as predicted by the literature.

Columns (2) and (3) show sub-sample analysis for female and male workers of equation (1). In this case, the married dummy is found to be significant and positive only for male workers. Furthermore, the father dummy remained significant and positively associated with income, while the mother dummy became insignificant.

Table 5.1: Mincer equations

	Outcome: log wages					
	Pooled OLS			Fixed-effects		
Variables	Whole sample (1)	Female (2)	Male (3)	Whole sample (4)	Female (5)	Male (6)
Female	-0.198***					
	(0.011)					
Mother	-0.005	-0.003		-0.104	-0.145**	
	(0.015)	(0.016)		(0.082)	(0.074)	
Married	0.078***	0.010	0.119***			
	(0.010)	(0.017)	(0.013)			
Number of children	0.007	0.017**	-0.002	-0.018	0.004	-0.023
	(0.005)	(0.008)	(0.007)	(0.022)	(0.031)	(0.029)
Father	0.077***		0.082***	0.038		0.046
	(0.013)		(0.014)	(0.075)		(0.082)
Job interruption	-0.298***	-0.285***	-0.267***	-0.068*	-0.255***	0.043
	(0.030)	(0.042)	(0.043)	(0.036)	(0.050)	(0.050)
Observations	28188	10528	17660	28188	10528	17660
R^2	0.419	0.482	0.395	0.052	0.048	0.068
Individual covariates	Yes	Yes	Yes	Yes	Yes	Yes
Household covariates	Yes	Yes	Yes	Yes	Yes	Yes
Supply-side covariates	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors reported in parenthesis. Columns (2), (3), and (5), (6) show estimations for female and male sub-samples, respectively. Individuals controls include job status, age, age squared, area of residence, marital status, educational attainment, industry, and a dummy variable for individuals studying or in training. Household controls include the number of children, household size, and dummies variables for households receiving remittances, pension, or social programs such as school meals. Supply-side controls include energy consumption and dummies for economic strata. Significance at 1 percent, at 5 percent, and at 10 percent are represented by ***, **, and *, respectively.

The fixed-effects model presented in column (5) shows that the changes in motherhood, i.e., becoming a mother, has a significant and negative effect on incomes, representing a reduction of around 15 percent. On the contrary, the changes in the father dummy are not significant (columna 6). Lastly, detachment of the labor force has a significant and negative effect only for women, implying on average, a 26 percent reduction of women's income when being out of the labor force, unemployed or inactive.

5.2. Oaxaca-Blinder decompositions

Decomposition methods are useful to further investigate differences in wages. The Oaxaca-Blinder method allows to decompose the wage differentials between men and women into an explained part due to differences in characteristics and an unexplained residual part. The effect of different productive characteristics is called the endowment effect, and the unexplained residual, the differences in the estimated coefficients for both groups, is often referred to as the "discrimination effect" (Weichselbaumer & Winter-Ebmer, 2005).

In the estimation of the Oaxaca-Blinder decomposition, I cannot control for possible threats to the method. For instance, differences in productivity measurement; employers might have a different way of assessing productivity than the one implemented in the model as observed characteristics. Likewise, the included observed characteristics might also be affected by "discrimination," and I do not adjust the estimations for this possible threat. Furthermore, I attempt to control for self-selection issues into the labor force by using Heckman's two-step correction while modeling labor force participation as a function of age, marital status and the number of children. However, this approach only addresses sample-selection issues driven by observed characteristics.

To investigate the possible triple penalty of women, motherhood and informality, I conduct Oaxaca-Blinder decompositions with and without Heckman's correction for three different subsamples. First, a sample that includes all men and women, second, a sample that includes only mother and father, and lastly, a sample that includes only mothers and fathers in the informal sector. The observed characteristics controlled for in the models are the same: age, educational attainment and industry at two-level digit. The major focus is not on the magnitudes of the gaps, but on the direction and significance of the effect that parenthood and informality have on the wage differentials. Several issues, such as measurement error in the outcome variable, could create misleading conclusions when interpreting magnitudes of the gap. However, I assumed no systematic difference in measuring the outcome variable across the different sub-samples analyzed.

For the decomposition without Heckman's correction, I use wages in the level to see raw differences. However, the definitive models with self-selection corrections have been estimated using the logarithm of wages as outcome variables.

As observed in all the different specifications, an unexplained wage differential exists. For the whole sample of men and women in the model without corrections, the total wage gap is around 44 percent, of which individual characteristics "endowments" contribute to its reduction. In the sample of mothers and fathers, the magnitude of the wage gap increases compared to the whole sample and endowments no further contribute to its reduction, indicating a possible penalty on motherhood.

Moreover, the decomposition analysis for mothers and fathers in the informal sector shows a wage gap reduction compared to mothers and fathers' whole sample. However, this reduction is mainly driven by an increase in the endowments' contribution. Therefore, the latter contributes to reducing the negative effect of being in the informal sector.

The decomposition analysis using sample-selection corrections for men and women shows similar results, with a contribution of endowments reducing the wage gap. In the case of the sample of mothers and fathers, a wage gap is still observed and it increases with respect to the wage gap of only women and men. Distinctly to the model without correction, endowments such as education level and industry contribute to reducing the wage gap between fathers and mothers. Further details of the Blinder-Oaxaca decompositions can be found in 5.2.

The decomposition results of the sample of fathers and mothers in the informal sector show an increase in the wage gap with respect to the total sample and the sample of only fathers and mothers. The contribution of endowments drives the changes after controlling for selection issues. Thus, informality implies an additional penalty to women after controlling for labor force participation.

Observed characteristics drive the changes in the correct and uncorrected models. For instance, the presence of educated women and mothers in the informal sector biases the estimations of the wage gap if not controlled for. Thus, being a mother and being in the informal sector represents additional penalties in terms of wage differentials compared to their male and father peers.

Table 5.2: Oaxaca-Blinder decomposition

	Oaxaca-Blinder decomposition Outcome: wages (córdobas)			Oaxaca-Blinder decompositionr with Heckman correction Outcome: log wages		
	Male vs. Female	Father vs. Mother	Father vs. Mother Informal sector	Male vs. Female	Father vs. Mother	Father vs. Mother Informal sector
	(1)	(2)	(3)	(4)	(5)	(6)
Male/father wage	2891.015***	3876.516***	3128.636***	8.538***	8.621***	8.492***
	(28.575)	(57.036)	(56.023)	(0.024)	(0.113)	(0.172)
Female/mother wage	1625.810***	1840.242***	1992.388***	7.534***	7.501***	7.311***
	(17.889)	(30.043)	(40.107)	(0.008)	(0.013)	(0.014)
Differences	1265.205***	2036.274***	1136.248***	1.004***	1.120***	1.181***
	(33.713)	(64.465)	(68.899)	(0.025)	(0.113)	(0.173)
Endowments	-108.674***	166.591***	-260.019***	-0.163***	-0.058***	0.044***
	(11.861)	(20.463)	(26.069)	(0.007)	(0.009)	(0.012)
Coefficients	1428.659***	1848.279***	1866.017***	1.171***	1.217***	1.351***
	(30.997)	(60.112)	(72.934)	(0.024)	(0.113)	(0.175)
Interaction	-54.780***	21.403	-469.750***	-0.004	-0.039***	-0.214***
	(6.466)	(14.762)	(40.659)	(0.005)	(0.007)	(0.015)
Observations	80863	28159	18015	45908	18159	13114

Notes: Standard errors reported in parenthesis. From column (1) to column (3) the outcome variable is wages in córdobas. From column (4) to column (6) the outcome is log wages. Significance at 1 percent, at 5 percent, and at 10 percent are represented by ***, **, and *, respectively.

5.3. Markov chain transition matrices

The question that arises is if those women and mothers are self-selecting themselves into this informal sector or if it is a result of the lack of formal jobs. Even though this question requires more exhaustive research and identification strategies, I intend to provide insights of the patterns of mobility of women and mothers compare with their peers in different employment statuses.

The Markov matrices' transition probabilities intend to approximate mobility and absorption of different employment statuses as the willingness or residual effect of labor market decisions. For instance, observing high mobility among those women who initially entered a formal job but later moving to an informal job or to unemployment or inactivity might indicate willingness and desire for more flexible jobs and hours. Nonetheless, these results are not causal and are merely the reflection of the patterns of individuals' mobility in the sample. Further details of the Markov Chain transition matrices can be found in 5.3.

5.3.1. Transitions by sex

Women initially observed in informal jobs have a 58 percent probability of remaining in this state, 15 percent of becoming informal own-account workers, and 16 percent of move out of the labor force. Men present similar patterns, but instead of moving out of the labor force, 25 percent transit from informal salaried to informal own-account.

On the other hand, formal salaried workers seem to be an absorbing state for women; once they enter a formal job, 91 percent of them remain in these jobs, only 3 percent move to informal salaried workers. Men on the contrary, once have entered a formal salaried position, 6 percent of them move to the informal salaried worker and 89 remain as formal. Similarly, informal own-account worker appears to be an absorbing status for men and women; 78 of men and 77 percent of women that enter those jobs remain on them.

Similarly, unemployment and inactive or out of the labor force have similar permanence probability for men and women; however, the type of jobs to which they transit afterward differs. Most of men moving from unemploy or inactive moved to informal salaried workers, while for women, 17 percent of unemployed and 20 percent of the inactive moved to informal own-account. This result might be in line with the observed differences in the informal sector observed in the Oaxaca-Blinder decomposition.

5.3.2. Transitions by parenthood

For mothers, similar patterns are observed in the formal salaried status, 91 percent of the mothers initially observed in this type of job remained there. For informal salaried workers, 19 percent transited to informal own-account and only 16 percent moved to unemployed or inactive, lower than the 21 percent observed for the whole sample.

Interestingly, the probability of remaining unemployed or inactive reduces for mothers compared with women no mother, with an increase in transitions towards informal salaried, but mostly towards informal own-account workers. Fathers also lower their probabilities of staying unemployed or inactive, now increasing the transitions towards own-account workers.

Formal jobs for women and mostly for mothers seem to be the absorbing and preferred status. Likewise, mothers transit-out of unemployment and inactivity more frequently compared with women's transitions with the whole sample.

Table 5.3: Markov chain transition matrices

	Salaried worker		Own-account worker		Unemployed	Out of the
	Informal	Formal	Informal	Formal		labor force
Job status	(1)	(2)	(3)	(4)	(5)	(6)
Period t	*Period $t+1$*					
Panel A. Male						
Informal salaried worker	61.1	4.0	25.3	1.3	4.0	4.3
Formal salaried worker	6.1	88.9	2.2	0.2	2.4	0.2
Informal own-account workerl	13.6	0.7	78.0	2.7	1.5	3.5
Formal own-account worker	14.1	0.8	48.4	25.0	1.6	10.2
Unemployed	32.7	13.0	14.3	1.0	22.9	16.2
Out of the labor force	11.6	0.9	17.3	1.5	6.3	62.5
Panel B. Female						
Informal salaried worker	58.2	4.1	15.4	1.8	4.9	15.6
Formal salaried worker	2.5	91.3	1.8	0.3	2.7	1.3
Informal own-account worker	5.6	0.5	76.6	2.8	1.8	12.8
Formal own-account worker	5.7	1.6	43.1	13.8	1.6	34.1
Unemployed	17.3	4.5	17.0	1.1	26.1	33.8
Out of the labor force	4.6	0.5	20.1	1.7	4.4	68.7
Panel C. Fathers						
Informal salaried worker	63.8	6.2	24.9	0.7	3.6	0.8
Formal salaried worker	5.6	89.7	3.1	0.2	1.1	0.2
Informal own-account worker	10.4	1.0	84.0	2.5	1.1	1.0
Formal own-account worker	15.8	0.0	55.3	23.7	2.6	2.6
Unemployed	32.4	9.9	31.0	2.8	15.5	8.5
Out of the labor force	10.7	1.0	26.2	2.9	6.8	52.4
Panel D. Mothers						
Informal salaried worker	59.1	3.7	19.2	2.3	3.6	12.2
Formal salaried worker	2.6	90.8	2.9	0.0	2.0	1.8
Informal own-account worker	5.6	0.3	79.4	2.8	1.0	11.0
Formal own-account worker	9.3	2.8	46.3	12.0	1.9	27.8
Unemployed	22.8	2.1	21.4	2.1	22.8	29.0
Out of the labor force	4.8	0.3	26.9	2.2	3.5	62.2

Notes: The sum of each row is equal to 100. Thus, each cell shows the percentage of individuals falling into each respective job status.

6. Conclusions

In this paper, I review existing literature on the gender wage gaps, trends, and potential explanations, focusing on the research that aimed to contribute to the study of the unexplained components of the wage differentials between men and women. Hence, a special emphasis on the reasons such as the family gap and informality was made. This study also intended to provide evidence of the existence of a gender wage gap in Nicaragua and the possible triple penalty that a mother in a country with high rates of informality might face.

To do so, I have employed methodologies commonly used by the available literature to shed light on the impact of gender on earnings, wage differentials among different groups, the impact of motherhood, and further information on the selection into flexible jobs for women and mothers.

The review of the most relevant and recent literature shows, on the one hand, that traditional factors, especially human capital-related, explain very little of the current gaps in earnings. On the other hand, it reveals that research on the impact of aspects such as tenure and labor force participation present mixed results depending on the labor market conditions and other characteristics. Furthermore, empirical evidence in the field of family gaps is also controversial due to endogeneity concerns. However, substantial and convincing research has recently shown the impact of motherhood on gender inequality. This research's primary focus has been on developed economies with very particular welfare states and labor market conditions.

The estimation of wage equations allowed to identify the effect of gender on earnings and additional variables that have essential roles in determining wages for women and men in Nicaragua. While fatherhood and marriage positively impact men's earnings, motherhood and marriage have likely adverse effects on women's wages. The number of children did not affect the mother penalty; therefore, we could suspect that the effects in earnings start from the first child and remains constant regardless of the number of children. Similarly, job interruptions in the previous quarter seem to have negative impacts on wages for women only.

Decomposition analysis showed the effect of gender, parenthood, formalization status and the role of women's unobserved characteristics when participating in the labor force. The presented evidence suggests the presence of a gender wage gap, even after correcting for self-selection into the labor force. Furthermore, the decomposition estimations by different subsamples, in this case between men and women as a whole and fathers and mothers, provide insights into the widener of the wage gap given parenthood status. Thus, a motherhood penalty is likely to exist, increasing the gap that women face during their working lives compared with their male peers and women without children.

Additionally, after correcting for self-selection, Oaxaca-Blinder decompositions provided further evidence of the impact of informality and the unobserved characteristics of the mothers participating in the sector. The final model suggests an additional penalty imposed by the informal sector that is counterweight with the mothers' human-capital characteristics participating in the informal sector.

The analysis presented by the transition matrices suggests that formal salaried jobs are a desirable and absorbing state for women and mothers. Women and mothers presented even higher staying probabilities than men and more mobility out of unemployment and inactivity. It argues that informal or more flexible jobs are not more desired by women and mothers but are the result of the lack of better jobs in the market. However, further evidence is needed to support this claim.

Although this study did not aim to analyze the nature of the discriminations, given the characteristics of the labor market and labor market mobility, statistical discrimination is likely to be present in this context. Furthermore, in a possible scenario of discriminatory behavior, employment protection legislation against gender discrimination could play a pivotal role in reducing the persistence of a gender wage gap. An important step would be improving the legal framework with an extensive and participatory review of successful legislation implemented in countries with similar conditions. Among this legislation, it is meritorious a particular emphasis on parental leave, its burden for the private sector, and its effects on the employment decisions and labor market attachment for women. Additional research is needed to identify the sources of discrimination and inform policymakers.

Finally, regarding a possible gender bias caused by the time allocation decisions for reproductive work, proper family policies such as childcare and parental leave are critical to counteract women's disadvantages in the formal market. Qualitative improvements in the childcare offer and the extension of parental leave for fathers constitute potential options to reduce women's disadvantages and employer's discrimination.

References

Altonji, J. G., & Blank, R. M. (1999). Race and gender in the labor market. In *Handbook of labor economics* (pp. 3143–3259). Elsevier. https://doi.org/10.1016/s1573-4463(99)30039-0

Angrist, J. D., & Pischke, J.-S. (2008). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.

Arrow, K. J. (1973). The theory of discrimination. In *Discrimination in labor markets* (pp. 1–33). Princeton University Press. https://doi.org/10.1515/9781400867066-003

Banco Central de Nicaragua. (2009). *50 años de estadísticas macroeconómicas 1960-2009*.

Becker, G. S. (1985). Human capital, effort, and the sexual division of labor. *Journal of Labor Economics*, *3*(1, Part 2), S33–S58. https://doi.org/10.1086/298075

Becker, G. S., Hubbard, W. H. J., & Murphy, K. M. (2010). The market for college graduates and the worldwide boom in higher education of women. *American Economic Review*, *100*(2), 229–233. https://doi.org/10.1257/aer.100.2.229

Benard, S., & Correll, S. J. (2010). Normative discrimination and the motherhood penalty. *Gender & Society*, *24*(5), 616–646. https://doi.org/10.1177/0891243210383142

Berniell, I., Berniell, L., de la Mata, D., Edo, M., & Marchionni, M. (2021). Gender gaps in labor informality: The motherhood effect. *Journal of Development Economics*, *150*, 102599. https://doi.org/10.1016/j.jdeveco.2020.102599

Blau, F., & Kahn, L. (1992). *The gender earnings gap: Some international evidence* (National Bureau of Economic Research No. 4224). https://doi.org/10.3386/w4224

Blau, F., & Kahn, L. (2000). Gender differences in pay. *Journal of Economic Perspectives*, *14*(4), 75–100. https://doi.org/10.1257/jep.14.4.75

Blau, F., & Kahn, L. (2017). The gender wage gap: Extent, trends, and explanations. *Journal of Economic Literature*, *55*(3), 789–865. https://doi.org/10.1257/jel.20160995

Blinder, A. S. (1973). Wage discrimination: Reduced form and structural estimates. *The Journal of Human Resources*, *8*(4), 436. https://doi.org/10.2307/144855

Borjas, G. (2016). *Labor economics* (7th). McGraw-Hill.

Bosch, M., & Maloney, W. F. (2010). Comparative analysis of labor market dynamics using markov processes: An application to informality. *Labour Economics*, *17*(4), 621–631. https://doi.org/10.1016/j.labeco.2010.01.005

Budig, M. J., & England, P. (2001). The wage penalty for motherhood. *American Sociological Review*, *66*(2), 204–225. https://doi.org/10.2307/2657415

Cha, Y., & Weeden, K. A. (2014). Overwork and the slow convergence in the gender gap in wages. *American Sociological Review*, *79*(3), 457–484. https://doi.org/10.1177/0003122414528936

Correll, S. J., Benard, S., & Paik, I. (2007). Getting a job: Is there a motherhood penalty? *American Journal of Sociology*, *112*(5), 1297–1339. https://doi.org/10.1086/511799

del Boca, D., Flinn, C., & Wiswall, M. (2013). Household choices and child development. *The Review of Economic Studies*, *81*(1), 137–185. https://doi.org/10.1093/restud/rdt026

Elias, P. (1997). *Occupational classification (isco-88): Concepts, methods, reliability, validity and cross-national comparability* (OECD Labour Market and Social Policy Occasional Papers No. 20). Organisation for Economic Cooperation; Development.

Fortin, N., Lemieux, T., & Firpo, S. (2011). Decomposition methods in economics. In *Handbook of labor economics* (pp. 1–102). Elsevier. https://doi.org/10.1016/s0169-7218(11)00407-2

Gaddis, I., & Klasen, S. (2013). Economic development, structural change, and women's labor force participation: *Journal of Population Economics*, *27*(3), 639–681. https://doi.org/10.1007/s00148-013-0488-2

Goldin, C. (2006). The quiet revolution that transformed women's employment, education, and family. *American Economic Review*, *96*(2), 1–21. https://doi.org/10.1257/000282806777212350

Goldin, C. (2008). Gender gap. *The Library of Economics and Liberty*.

Goldin, C. (2014). A grand gender convergence: Its last chapter. *American Economic Review*, *104*(4), 1091–1119. https://doi.org/10.1257/aer.104.4.1091

Heckman, J. J. (1979). Sample selection bias as a specification error. *Econometrica*, *47*(1), 153. https://doi.org/10.2307/1912352

Hersch, J., & Stratton, L. S. (2002). Housework and wages. *The Journal of Human Resources*, *37*(1), 217. https://doi.org/10.2307/3069609

International Labour Organization. (2013). *Measuring informality: A statistical manual on the informal sector and informal employment*.

International Labour Organization. (2018). *Global wage report 2018/19: What lies behind gender pay gaps*.

Jann, B. (2008). The blinder–oaxaca decomposition for linear regression models. *The Stata Journal: Promoting communications on statistics and Stata*, *8*(4), 453–479. https://doi.org/10.1177/1536867x0800800401

Kleven, H., Landais, C., & Søgaard, J. E. (2019). Children and gender inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, *11*(4), 181–209. https://doi.org/10.1257/app.20180010

Maloney, W. F. (2004). Informality revisited. *World Development*, *32*(7), 1159–1178. https://doi.org/10.1016/j.worlddev.2004.01.008

Mincer, J., & Polachek, S. (1974). Family investments in human capital: Earnings of women. *Journal of Political Economy*, *82*(2, Part 2), S76–S108. https://doi.org/10.1086/260293

Monroy, E. (2008). *Equidad de género en el mercado laboral Nicaragua* (Serie Cuadernos de Género para Nicaragua No. 1). Banco Mundial. Washington, D.C.

Nichols, A. (2014). Measuring mobility. *Stata Conference 11, Stata Users Group*.

Nordman, C. J., Rakotomanana, F., & Roubaud, F. (2016). Informal versus formal: A panel data analysis of earnings gaps in madagascar. *World Development*, *86*, 1–17. https://doi.org/10.1016/j.worlddev.2016.05.006

Oaxaca, R. (1973). Male-female wage differentials in urban labor markets. *International Economic Review*, *14*(3), 693. https://doi.org/10.2307/2525981

Phelps, E. (1972). The statistical theory of racism and sexism. *American Economic Review*, *62*(4), 659–661.

Programa de las Naciones Unidas para el Desarrollo. (2014). *El mercado laboral de Nicaragua desde un enfoque de género* (Cuaderno de Desarrollo Humano No. 6). Programa de las Naciones Unidas para el Desarrollo. Managua.

Ruhm, C. J. (1998). The economic consequences of parental leave mandates: Lessons from europe. *The Quarterly Journal of Economics*, *113*(1), 285–317. https://doi.org/10.1162/003355398555586

Ruzik, A., & Rokicka, M. (2010). The gender pay gap in informal employment in poland. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1674939

Sigle-Rushton, W., & Waldfogel, J. (2007). The incomes of families with children: A cross-national comparison. *Journal of European Social Policy*, *17*(4), 299–318. https://doi.org/10.1177/0958928707082474

Tansel, A., Keskin, H. I., & Ozdemir, Z. A. (2019). Is there an informal employment wage penalty in egypt? evidence from quantile regression on panel data. *Empirical Economics*, *58*(6), 2949–2979. https://doi.org/10.1007/s00181-019-01651-2

United Nations. (2008). *International standard industrial classification of all economic activities (isic)*.

Waldfogel, J. (1998). Understanding the 'family gap' in pay for women with children. *Journal of Economic Perspectives*, *12*(1), 137–156. https://doi.org/10.1257/jep.12.1.137

Weichselbaumer, D., & Winter-Ebmer, R. (2005). A meta-analysis of the international gender wage gap. *Journal of Economic Surveys*, *19*(3), 479–511. https://doi.org/10.1111/j.0950-0804.2005.00256.x

Williams, W. M., & Ceci, S. J. (2015). National hiring experiments reveal 2:1 faculty preference for women on STEM tenure track. *Proceedings of the National Academy of Sciences*, *112*(17), 5360–5365. https://doi.org/10.1073/pnas.1418878112

World Bank. (2012). *World development report 2012 : Gender equality and development*. World Bank. https://openknowledge.worldbank.org/handle/10986/4391

Yahmed, S. B. (2018). Formal but less equal. gender wage gaps in formal and informal jobs in urban brazil. *World Development*, *101*, 73–87. https://doi.org/10.1016/j.worlddev.2017.08.012